Java From Scratch

Lesson 1 Notes — Basics & Syntax

Level: Complete beginner

These notes cover everything learned so far: what Java is at a basic level, the structure of a Java program, variables, data types, printing, comments, braces, semicolons, case sensitivity, and the most important syntax symbols.

> Learning goal: By the end of these notes, you should be able to read and write a simple Java program and understand the basic syntax used in it.

Topics covered

1	First Java Program
2	Basic Java Program Structure
3	Statements and Semicolons
4	Case Sensitivity
5	Variables
6	Basic Data Types
7	Printing with System.out.println()
8	Comments
9	Braces and Code Blocks
10	The Three Important Syntax Symbols
11	Complete Example
12	Practice Questions & Quick Revision

1. Your First Java Program

A Java program is written using Java syntax. A simple program that prints a message looks like this:

```
public class Main {
public static void main(String[] args) {
System.out.println("Hello, World!");
}
}
```

Output:

```
Hello, World!
```

This is the traditional first example used when learning a programming language. For now, you do not need to memorize every word in the program. You mainly need to recognize the structure.

Important: Java starts running your program from the main() method.

2. Basic Java Program Structure

A beginner-friendly way to think about the structure is:

```
public class Main {
public static void main(String[] args) {

// Your code goes here

}
}
```

class Main — The program is placed inside a class named Main.

main() — This is the starting point of execution for a basic Java application.

{ } — Curly braces define blocks of code.

System.out.println() — Used to display information on the screen.

The words public, static, and void, as well as String[] args, will be explained later. At this stage, focus on recognizing the pattern.

3. Statements and Semicolons

A statement is an instruction that tells Java to do something. Most Java statements end with a semicolon ;.

```
int age = 20;
System.out.println("Hello");
```

The semicolon tells Java that the statement is finished.

Incorrect:

```
int age = 20
```

Correct:

```
int age = 20;
```

Forgetting a semicolon is a common beginner error and usually causes a compilation error.

4. Java Is Case-Sensitive

Java treats uppercase and lowercase letters as different characters. This is called case sensitivity.

```
age
Age
AGE
```

These are three different names to Java.

```
int age = 20;

System.out.println(age); // correct
System.out.println(Age); // error
```

If you create a variable called age, you must use exactly age when referring to it.

> Tip: Pay close attention to capitalization when writing Java code.

5. Variables

A variable is a named storage location used to keep a value. You can think of it as a labeled box that contains information.

General syntax:

```
dataType variableName = value;
```

Examples:

```
int age = 20;
double price = 99.50;
char grade = 'A';
boolean passed = true;
String name = "Alex";
```

The three main parts are:

Part	Meaning	Example
dataType	What kind of value is stored	int
variableName	The name you give the variable	age
value	The actual data stored	20

6. Basic Data Types

A data type tells Java what kind of data a variable can store.

Type	Example	Used for
int	25	Whole numbers
double	3.14	Decimal numbers
char	'A'	One character
boolean	true	True / false values
String	"Java"	Text

char vs String:

```
char letter = 'J';
String language = "Java";
```

A char stores a single character and uses single quotes. A String stores text and uses double quotes.

7. Printing with System.out.println()

Use System.out.println() to print information to the console.

```
System.out.println("Hello");
System.out.println(20);
```

You can also print the value stored in a variable:

```
int age = 20;
System.out.println(age);
```

You can combine text and variables using the + operator:

```
String name = "Alex";
int age = 20;

System.out.println("Name: " + name);
System.out.println("Age: " + age);
```

Here, + joins text and a value together to form the output.

8. Comments

Comments are notes written inside code for humans. Java ignores comments when executing the program.

Single-line comment:

```
// This is a comment
int age = 20;
```

Multi-line comment:

```
/*
This is
a multi-line comment
*/
```

Comments are useful for explaining code, leaving reminders, or temporarily describing a section of a program.

9. Braces and Code Blocks

Curly braces { } are used to group statements into a block of code.

```
if (age >= 18) {
System.out.println("Adult");
}
```

The statements between { and } belong to that block.

You will use braces frequently with classes, methods, if/else, loops, and other Java structures.

10. The Three Most Important Syntax Symbols

At this stage, remember these three symbols first:

Symbol	Meaning	Example
;	Ends a statement	int age = 20;
()	Used for conditions and method parameters	main(String[] args)
{}	Defines a block of code	if (...) { ... }

Easy memory trick:

; → finish an instruction
() → put parameters or a condition here
{} → put a block of code here

11. Complete Example

The following program combines the concepts covered so far:

```
public class Main {
public static void main(String[] args) {

String name = "Sam";
int age = 20;
int favoriteNumber = 7;

System.out.println("My name is " + name);
System.out.println("I am " + age + " years old");
System.out.println("My favorite number is " + favoriteNumber);
}
}
```

Output:

```
My name is Sam
I am 20 years old
My favorite number is 7
```

What this example demonstrates:

- A class named Main
- The main() method
- Variables and data types
- String and integer values
- System.out.println()
- String concatenation using +
- Semicolons and curly braces

12. Quick Revision

What is syntax? Syntax is the set of rules that determines how Java code must be written so that Java can understand it.

What is a variable? A named place used to store a value.

What does int store? Whole numbers.

What does double store? Decimal numbers.

What does char store? A single character.

What does boolean store? true or false.

What does String store? Text.

How do you print? Use System.out.println().

Are Java programs case-sensitive? Yes.

What usually ends a Java statement? A semicolon ;.

What do { } do? They define a block of code.

What do ( ) do? They are commonly used for method parameters and conditions.

Practice Questions

Try answering these without looking back at the notes:

1. What is the general syntax for declaring a variable?
2. What is the difference between char and String?
3. Why is Age different from age in Java?
4. What symbol usually ends a statement?
5. What method is used to print output?
6. What do curly braces represent?
7. Write a program that stores your name and age and prints both.

Next lesson: Operators and Expressions — arithmetic operators, assignment, comparison, logical operators, and how Java evaluates expressions.

End of Lesson 1